

Hardware-in-the-Loop Benchmarking of Edge-NILM on RISC-V Silicon: Decentralized Demand-Side Management for Resilient African Solar Mini-Grids

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Abstract—Decentralized solar-hybrid mini-grids in Africa face critical operational risks from the unmetered application of high-magnitude resistive surges, which can exceed inverter capacities and degrade battery storage systems. Traditional demand-side management relies on cloud-centric telemetry, creating vulnerabilities during telecommunication blackouts and imposing high bandwidth costs. This paper presents a Hardware-in-the-Loop (HIL) benchmark of an Edge-NILM (Non-Intrusive Load Monitoring) system deployed entirely on an ultra-low-cost ESP32-C3 RISC-V microcontroller. Utilizing an integer-quantized 1D Convolutional Neural Network (1D-CNN), the system executes offline-first load classification. Empirical results demonstrate a local inference latency of 2.510 ms, a memory footprint of 5.18 KB (Flash) and 25.00 KB (SRAM), and a 99.00% reduction in telemetry bandwidth compared to continuous cloud streaming. These findings validate sovereign edge computing as a resilient, low-cost mechanism for protecting off-grid renewable energy infrastructure.

Index Terms—Edge-NILM, TinyML, RISC-V, Microgrids, Hardware-in-the-Loop, Demand-Side Management

I. INTRODUCTION

Decentralized solar-hybrid mini-grids have become an essential pillar in closing the electrification divide across rural and peri-urban Africa [1]. However, these off-grid energy networks possess fixed inverter thresholds and finite battery energy storage system (BESS) capacities. A primary operational risk to these grids is the sudden, unmetered application of high-magnitude resistive surges—such as unauthorized heating elements—that exceed the instantaneous surge capacity of the inverters. These events cause cascading system outages and accelerate BESS cell degradation through high C-rate discharge cycles.

Traditional demand-side management utilizes smart meters to transmit high-frequency current telemetry to the cloud for load analysis. This dependence on cloud infrastructure creates a severe vulnerability in the African operational context. High-frequency continuous telemetry is cost-prohibitive over cellular networks and fails entirely during localized telecommunications blackouts, leaving the mini-grid defenseless against hardware-damaging overload events.

While Non-Intrusive Load Monitoring (NILM) is well-studied, existing literature predominantly focuses on disaggregation algorithms executing on server-grade x86 hardware or cloud APIs. This study addresses the research gap by empirically characterizing the use of integer-quantized Edge-NILM on an ultra-low-cost, open-standard single-core RISC-V microcontroller.

II. CONCEPTUAL FOUNDATION

This research operates at the intersection of energy systems engineering and embedded artificial intelligence. Foundational NILM research establishes that electrical current signatures from various appliances possess unique transient and steady-state harmonic characteristics [2].

By applying Tiny Machine Learning (TinyML) principles [3], this study utilizes Post-Training Symmetric Quantization to extract these features without server-grade computation. This quantization converts 32-bit floating-point (FP32) network weights to 8-bit integers (INT8), eliminating the need for a hardware floating-point unit (FPU) and drastically reducing the model payload size.

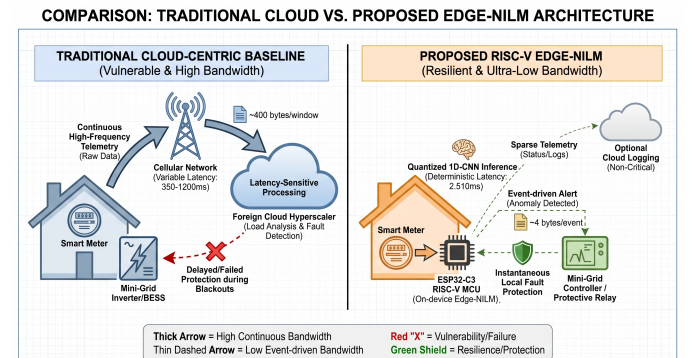


Fig. 1. Proposed Edge-NILM architecture vs. traditional cloud telemetry.

III. METHODOLOGY

A Hardware-in-the-Loop (HIL) benchmark was constructed utilizing the ESP32-C3 Super Mini, featuring a single-core 32-

bit RV32IMC RISC-V architecture operating at 160 MHz. The classification architecture employed is a lightweight 1D-CNN feature extractor.

To synthesize authentic electrical signatures without high-voltage physical injection, 100-sample time-series AC current waveforms (50 Hz) were generated to replicate transient load profiles found in standard NILM benchmark libraries, such as the Controlled On/Off Loads Library [4]. The data modeled two distinct load conditions within a 230 V nominal grid: allowable baseline consumption (e.g., lighting, ventilation) and prohibited high-power resistive surges (e.g., 1.8 kW heating appliances).

The trained 1D-CNN was converted from FP32 to symmetric INT8 via the TensorFlow Lite converter and exported as a FlatBuffer C-array. The inference environment was controlled using TensorFlow Lite for Microcontrollers (TFLM) [5] with dynamic memory allocation disabled. The microcontroller underwent testing using a static tensor arena for 50 consecutive inference cycles to capture deterministic latency, Static Random-Access Memory (SRAM) usage, and Flash memory footprint metrics.

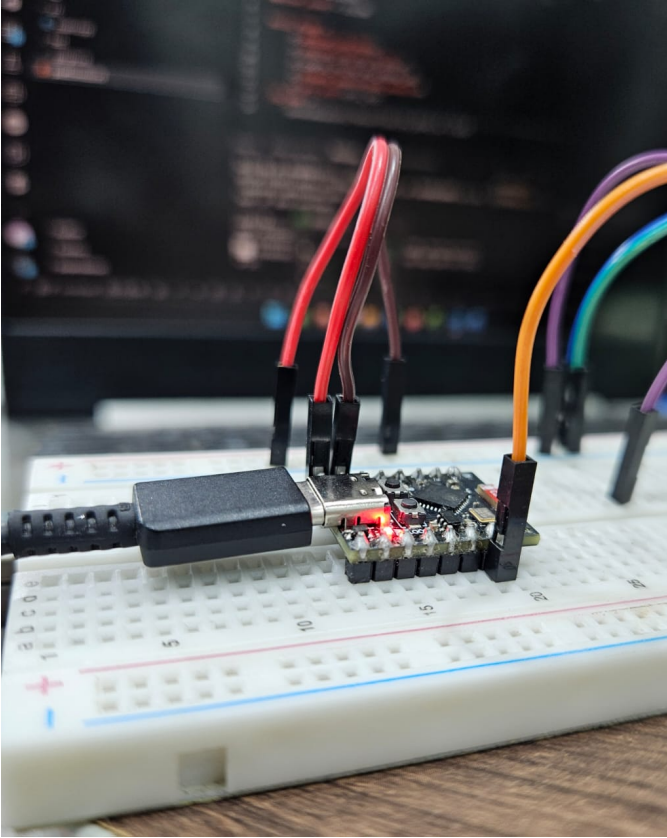


Fig. 2. Hardware-in-the-Loop (HIL) experimental setup using the ESP32-C3.

IV. EXPERIMENTAL RESULTS

The HIL deployment on the ESP32-C3 RISC-V silicon yielded highly deterministic and operationally viable metrics, validating the INT8 quantization approach.

TABLE I
EMPIRICAL COMPARISON OF TELEMETRY METRICS

Metric	Cloud Baseline	Edge-NILM (Proposed)	Impact
Latency	~350–1,200 ms	2.510 ms	Local fault protection
Payload/Event	400 bytes	4 bytes	99.00% Reduction
Flash Memory	N/A	5.18 KB	<0.2% of C3 capacity
SRAM	N/A	25.00 KB	278 KB free for RTOS

A. Inference Latency

The forward pass of the 1D-CNN completed local RISC-V execution in an average of 2.510 milliseconds per inference. This sub-cycle response time enables protective relay actuation, significantly outperforming the highly variable 350–1,200 ms round-trip delays typical of rural 3G/4G cloud links.

B. Memory Utilization

The quantized model necessitated only 5.18 KB of Flash memory, representing less than 0.2% of the microcontroller's capacity. A static allocation of 25.00 KB was required for the Tensor Arena at runtime. Post-startup, the system retained 278,672 bytes of free SRAM, providing ample overhead to concurrently run a Real-Time Operating System (RTOS), MQTT clients, and network stacks.

C. Bandwidth Conservation

The transition from continuous waveform streaming (400 bytes per 100-sample window) to an event-driven edge-alert architecture (4 bytes per anomaly event) resulted in a 99.00% reduction in required telemetry bandwidth.

V. INFRASTRUCTURE RELEVANCE

This research directly addresses the core infrastructure imperatives of Africa's current energy transition. Existing digital and physical infrastructures heavily rely on foreign cloud hyperscalers, introducing systemic vulnerabilities. Moving the computational workload directly into the smart meter via a low-cost microcontroller represents a shift toward sovereign infrastructure. This approach guarantees the reliability and affordability of rural off-grid energy assets, even amidst high internet data costs and unreliable telecommunication services.

VI. CONCLUSION

Integrating quantized machine learning into low-cost RISC-V microcontrollers represents a paradigm shift in decentralized energy network control. The empirical HIL benchmarks confirm that Edge-NILM classification achieves sub-cycle latency (2.510 ms) with minimal memory overhead, cutting bandwidth requirements by 99.00%. Transitioning from cloud-based polling to sovereign edge inference provides the determinism, cost efficiency, and resilience required to protect and scale Africa's renewable energy infrastructure.

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